

5TH INTERNATIONAL CONFERENCE

Smart Grid Synchronized Measurements & *Analytics*



Benchmarking Dynamic Frequency Estimators for Low-Inertia IBR Grids

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Date

June 1-4, 2026

Venue

Santiago, Chile



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Title

Say:

- Good morning. I am Jorge Mayorga.
- This talk benchmarks dynamic frequency estimators for low-inertia, inverter-based grids.
- The question: in a grid run by power electronics, which estimator do we trust?

Key line: The answer depends on what you are measuring.



Motivation

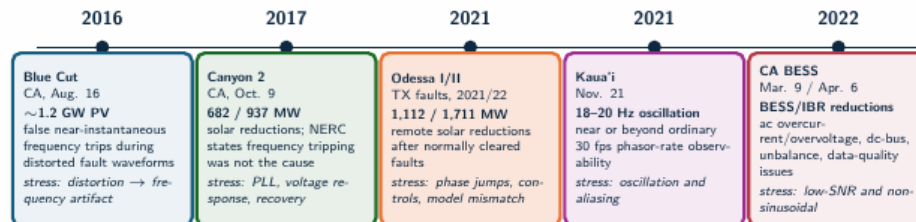
Why frequency estimation in low-inertia IBR grids is hard

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Logos: PES, IAS, cigre, ics

Key line: Let me start with why this matters.

Field events define the benchmark stress model



Benchmark implication: the estimator must be tested against waveform distortion, phase jumps, control recovery, unbalance, noise, harmonics, interharmonics, and reporting-rate limits, not only steady-state frequency error.

Source: NERC Blue Cut 2017; NERC/WECC Canyon 2 2018; NERC Odessa 2021/2022; NERC/WECC CA BESS 2023; Dong et al. Kauai analysis

Field events define the stress model

Say:

- Real IBR disturbances are not clean sine waves.
- Blue Cut, Odessa, the Kauai oscillation: phase jumps, fast RoCoF, harmonics and noise stacked together.
- We use these real events to define the stress cases the benchmark must cover.

Key line: Field events, not textbook signals, set our test bar.

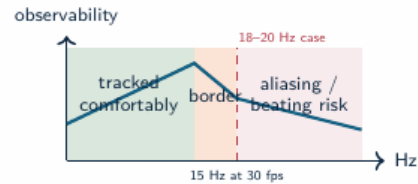
Standard PMU streams show consequences, not always causes

Fast fault and ride-through

IBR controls can react within cycles. A low-rate or heavily filtered frequency estimate may miss the subcycle cause while still showing the delayed consequence.

Oscillation near reporting-rate limits

The Kaua'i Island event on Nov. 21, 2021 exhibited 18–20 Hz oscillations after an oil-power-plant trip. A 30 frame/s phasor stream has a 15 Hz Nyquist limit for phasor-time variations, so oscillations near this range can alias or appear as beating.



Harmonics and interharmonics

Standard synchrophasor outputs summarize the fundamental component; harmonic and interharmonic contamination can still dominate estimator failure modes.

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PMU streams show consequences

Say:

- Standard PMU streams report frequency and RoCoF.
- During IBR events those numbers show the consequence, not always the cause.
- The estimator inside the PMU shapes what we see.

Key line: Choosing the estimator is itself a measurement decision.

Research problem

- Low-inertia IBR grids require reliable local frequency estimates during ramps, phase jumps, harmonic distortion, ringdown, and mixed dynamic events.
- Classical estimators often trade off responsiveness, noise rejection, latency, and computational load.
- Modern model-based and data-driven estimators improve some regimes, but can be difficult to compare fairly.
- Existing comparisons often under-specify seeds, disturbance families, tuning policy, trip-risk, structural latency, or computational cost.

Research gap

The literature needs a reproducible benchmark that connects estimator error to protection-oriented risk, settling behavior, and deployability.

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Research problem

Say:

- There is no agreed way to compare estimators under IBR-specific stress.
- Compliance tests use clean signals.
- They do not tell us how a method behaves on a composite event.

Key line: The gap is a fair, IBR-relevant comparison.

Operating claim

- Frequency estimation under grid disturbances is a **multi-objective measurement problem**.
- Rankings change with the objective: frequency RMSE, RoCoF error, latency, CPU cost, and disturbance sensitivity produce different orders.
- IBR-era disturbances make this operational: distorted waveforms and fast controls can turn a measurement artifact into a ride-through or forensic diagnosis problem.
- Estimator choice is conditional: event family, objective metric, and compute budget determine the preferred method.

Measurement rule

Report error, transient exposure, settling, latency, and cost as separate outputs for each estimator.

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Operating claim

Say:

- Estimator selection must be conditional.
- It depends on the disturbance type, the metric you care about, and the compute budget.

Key line: There is no single best estimator.

Technical contributions

Not claimed here

- PMU time-stamp certification;
- a single universal leaderboard;
- hardware latency measurements.

Contributions

- OpenFreqBench: a reproducible stress-test benchmark and platform for relay-class estimators;
- a metric-complete comparison protocol: RMSE, peak error, RoCoF error, T_{trip} , settling, and CPU;
- evidence that standard compliance tests miss IBR failure modes;
- an adapted RA-EKF as a tested estimator, with explicit $\hat{\omega}$ state and event gating.

The benchmark explains why estimator rankings change across objectives, regimes, and compute budgets.

Technical contributions

Say:

- One: an open benchmark platform with known-truth scenarios.
- Two: a common execution contract for 18 estimators.
- Three: a metric vector separating accuracy, trip-risk, latency and cost.

Key line: Open platform, common contract, multi-metric evidence.

Experiment scale

18
estimators

5
scenarios

4
families

grid
search

Central finding

No single estimator wins across objectives: the ranking changes with scenario, metric, and compute budget. Protection-class frequency measurement needs tests beyond compliance-style cases.

Dataset: 18 estimators (incl. legacy baselines) across 33 scenarios, $n = 5$ Monte Carlo per scenario, with standard and IBR-specific metrics, plus ATLAS severity sweeps.

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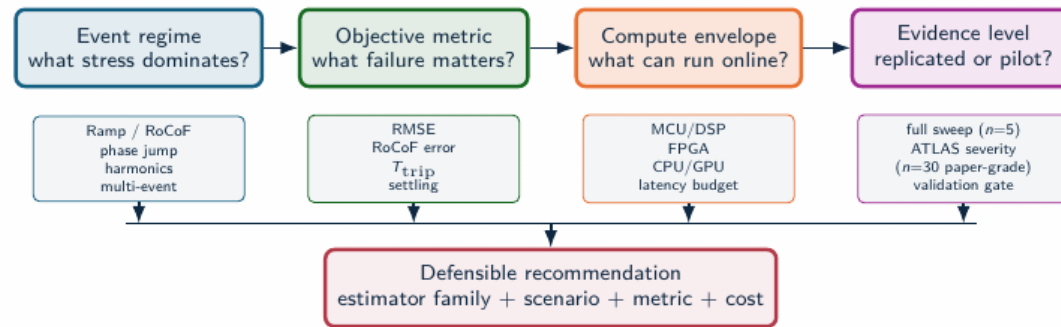
Experiment scale

Say:

- 18 estimators, 33 scenarios, five families of stress.
- Monte-Carlo repetitions, plus a severity-sweep engine we call ATLAS.
- Everything is reproducible from artifacts.

Key line: Large, factorial, and fully reproducible.

Estimator selection logic



Example: RA-EKF is recommended for ramp/phase-jump protection metrics under DSP/FPGA-class cost; global RMSE leadership remains scenario-dependent.

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Estimator selection logic (build-up)

Say:

- Selection follows three questions: which event class, which metric, which compute budget.
- Each one can point to a different method.

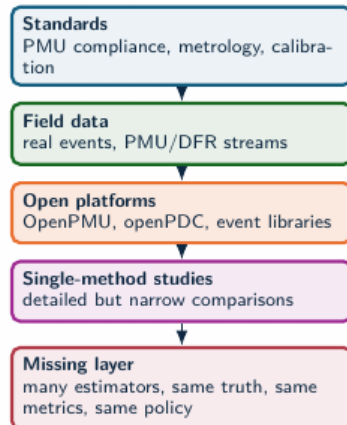
Key line: This diagram is the whole talk in one picture.

Benchmark and platform (section)

Key line: First, the benchmark and the platform.

The slide features a dark blue background with a geometric pattern of overlapping triangles and lines. In the top right corner, there is a small blue logo for SGSMA 2026. The main title "Benchmark & platform" is centered in a large, white, serif font. Below it, the subtitle "What is missing in prior work — and OpenFreqBench" is centered in a smaller, white, sans-serif font. At the bottom left, the text "SGSMA 2026 · SANTIAGO DE CHILE" is displayed in white. Along the bottom edge, there are four logos: PES (Power & Energy Society), IAS (Instrumentation & Measurement Society), cigre (United Utilities), and ics (International Council on Large Electric Systems).

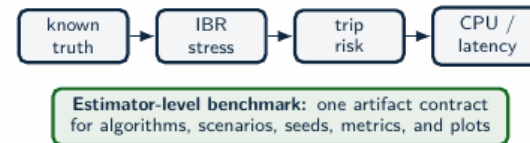
Benchmark literature gap



Result: controlled synthetic truth isolates estimator behavior under reproducible stress while field events define realism.

Why existing evidence is not enough

Standards define requirements; field events define realism; open PMU ecosystems provide infrastructure. None alone answers which estimator family is defensible under the same disturbance, tuning policy, metric vector, and compute budget.



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Benchmark literature gap

Say:

- Prior work compares a few estimators on one or two cases.
- No benchmark covers the full estimator space against composite IBR stress with reproducible metrics.

Key line: That is the gap we fill.

Benchmark comparison axes

Evidence source	Known truth	IBR composite stress	Trip-risk	CPU/latency	Reproducible artifacts
IEC/IEEE-style compliance	Yes	Limited	No	No	Device-dependent
Field PMU/DFR data	No	Yes	Possible	No	Dataset-dependent
Single-method studies	Often	Partial	Partial	Partial	Often limited
Estimator benchmark	Yes	Yes	Yes	Yes	Yes

Measured outputs

The same voltage trace is scored by RMSE, peak error, RoCoF error, trip exposure, settling, CPU, and structural latency.

Interpretive effect

An estimator can pass standard cases and still fail a composite IBR event; it can also be accurate but too delayed or costly for protection-adjacent use.

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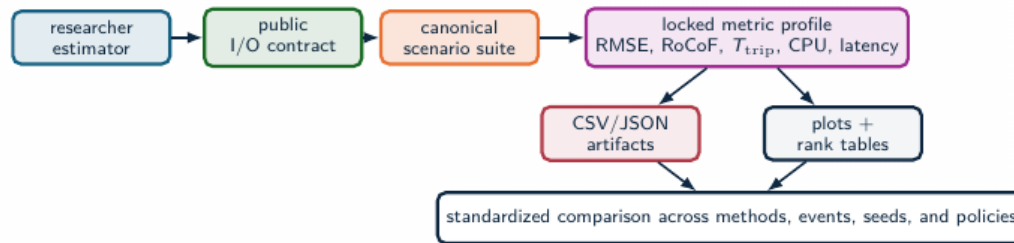
Benchmark comparison axes

Say:

- We compare on four axes: accuracy, robustness, computational cost, and trip exposure.
- A method can win one axis and lose another.

Key line: Four axes, because one number hides the trade-off.

OpenFreqBench platform



Researchers add

Estimator code, parameters, and hypotheses through the public contract.

Platform fixes

Truth signals, sampling, seeds, tuning policy, metric formulas, and schema.

Outputs

Benchmark reports, metric tables, plots, manifests, and readiness gates.

OpenFreqBench turns a new estimator into a repeatable benchmark entry instead of a one-off comparison.

Say:

- A scenario factory with known truth.
- A uniform estimator API and a locked metric engine.
- Traceable CSV and JSON artifacts.

Key line: Known truth in, locked metrics out, everything traced.

Experimental method

Same truth, same metrics, fair tuning

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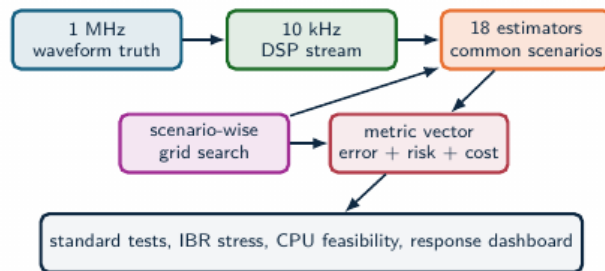
Experimental method (1)

Say:

- Every estimator sees the identical input trace.
- Same random seeds, same warm-up handling.

Key line: Fairness is enforced by construction.

Experimental method



Source: Methodology and test framework

Measurement setting

Local, non-time-synchronized relay-class frequency tracking from a decimated 10 kHz stream; PMU timestamp certification is outside scope.

Experimental principle

Each estimator–scenario pair is tuned within a declared grid, then compared using RMSE, peak error, trip exposure, settling, and CPU.

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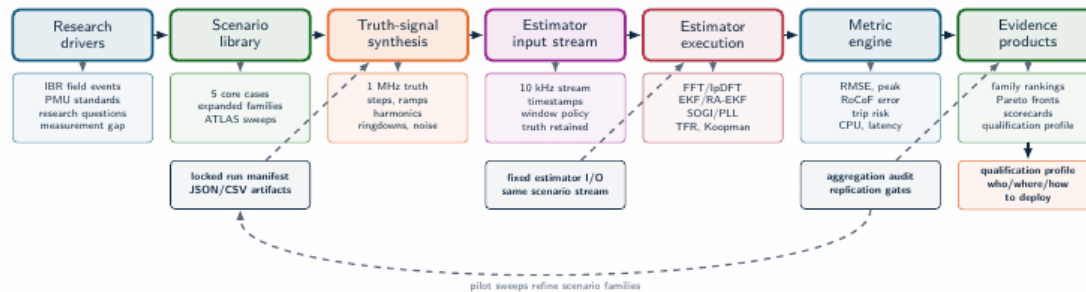
Experimental method (2)

Say:

- We tune under a fixed policy and exclude the warm-up region.
- CPU is measured as a response variable.

Key line: Nothing is hand-picked per method.

Full benchmark workflow



The benchmark is an evidence pipeline, not just a test script: every figure traces back to a locked manifest, a fixed I/O contract, and the readiness gate.

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Full benchmark workflow

Say:

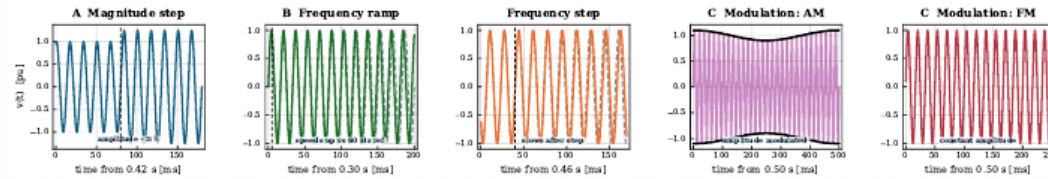
- Scenario, Monte-Carlo, estimator, locked metrics, then artifacts.
- Every figure in this talk traces back to those artifacts.

Key line: One pipeline, end to end, reproducible.

Standard stresses: the waveform is controlled

CORE SCENARIO SUITE

Scenario suite I — standard stresses: voltage $v(t)$ (top) vs. true frequency $f(t)$ (bottom)



Each standard test changes one physical ingredient at a time: magnitude, ramp, frequency step, AM, or FM.

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Standard stresses: controlled waveform

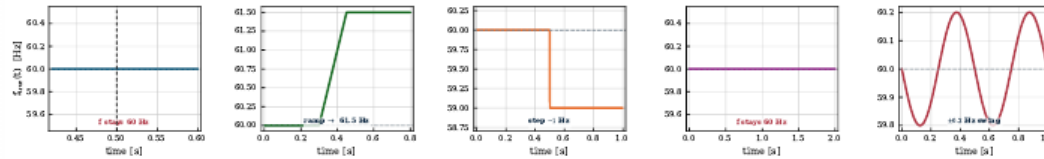
Say:

- The standard stresses are step, ramp, and modulation.
- The waveform is fully controlled, so we know the true frequency at every sample.

Key line: Known truth lets us score error exactly.

Standard stresses: the frequency truth is simple

CORE SCENARIO SUITE



The clean reference traces make estimator failure visible: divergence is an algorithmic response, not ambiguity in ground truth.

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Scenario suite I (dashboard)

Say:

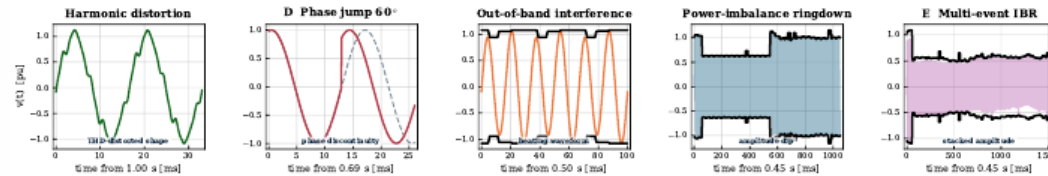
- Voltage on top, true frequency below, one column per scenario.
- For the magnitude step and AM, frequency stays flat at 60 hertz.

Key line: The stress is not always a frequency event, but estimators still react.

Composite IBR stresses: waveform physics are stacked

IBR STRESS SUITE

Scenario suite II — composite IBR stresses: voltage $v(t)$ (top) vs. true frequency $f(t)$ (bottom)



Composite cases combine harmonic distortion, phase jumps, out-of-band tones, ringdown recovery, and multi-event amplitude dynamics.

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Composite IBR stresses

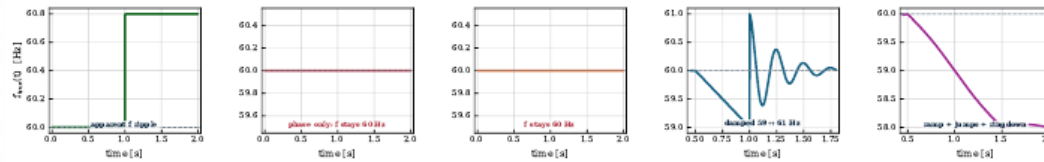
Say:

- Harmonics, phase jumps, RoCoF segments and ringdown, stacked into one waveform.
- This is what a real inverter-dominated fault produces.

Key line: Composite stress is the realistic case.

Composite IBR stresses: frequency truth becomes diagnostic

IBR STRESS SUITE



The estimator must recover through nonstationary segments, not just report low steady-state error after the event.

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Scenario suite II (dashboard)

Say:

- The composite suite, again voltage on top and frequency below.
- These are deliberately harder than any compliance test.

Key line: If a method survives here, it is robust.

Estimator families

Loop-based

PLL, SOGI-PLL, SOGI-FLL, Type-3 SOGI-PLL.

Model-based / stochastic

EKF, UKF, RA-EKF, LKF, LKF2.

Legacy baselines

TKEO and ZCD are retained as comparison references outside the recommended deployment set.

Window / spectral

IpDFT, TFT, ESPRIT, Prony, MUSIC.

Adaptive and data-driven

RLS, Koopman (RK-DPMU).

Interpretation rule

These families occupy different latency and stress-sensitivity positions. The comparison tracks where each family wins, degrades, and fails.

LKF and LKF2 follow H. Ahmed, S. Biricik, M. Benbouzid, "Linear Kalman Filter-Based Grid Synchronization Technique: An Alternative Implementation," IEEE, 2021.

Estimator families

Say:

- Loop-based PLLs, window and spectral methods, model-based Kalman filters, adaptive, and data-driven.
- We analyze by family because the family explains the behavior.

Key line: Eighteen estimators, five families.

Metrics and decision variables

Error and recovery

$$e[k] = \hat{f}[k] - f_{\text{true}}[k]$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_k e[k]^2}$$

$$E_{\text{max}} = \max_k |e[k]|$$

Protection-oriented risk

$$T_{\text{trip}} = \Delta t \sum_k \mathbf{1}\{|e[k]| > 0.5 \text{ Hz}\}$$

$$t_s = \min\{t : |e[\tau]| \leq \epsilon, \forall \tau \geq t\}$$

$$C_{\text{CPU}} = \text{mean time/sample}$$

The 0.5 Hz threshold is a protection-oriented risk proxy. The metric set separates average error, transient severity, trip exposure, settling, and compute cost.

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Metrics and decision variables

Say:

- RMSE for average error, peak error, and a trip-risk time.
- Trip-risk counts how long the error exceeds half a hertz; plus settling and CPU.

Key line: Half a hertz is our protection-oriented proxy.

Core evidence (section)

Key line: Now the core evidence: five scenarios, standard to composite.

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Standard scenario results

Method	Step	Ramp	Mod.
RA-EKF	0.000	0.006	0.037
SOGI-FLL	0.005	0.013	0.040
Koopman	0.003	0.017	0.044
TFT	0.001	0.019	0.046
UKF	0.000	0.017	0.055
LKF2	0.138	0.015	0.028
PLL	0.033	1.47	0.227
IpDFT	0.000	14.4	0.114
EKF	0.000	37.1	0.064

RMSE [Hz], mean over 5 seeds. Green:

below 0.05 Hz guide; red: failure/divergence regime.

What the heatmap shows

The standard step is easy for most methods. The ramp is the separator: EKF and IpDFT can look excellent at a point and still diverge in another seed.

Operational consequence

Robustness is not the same as point accuracy. A standard scenario can pass while the same estimator remains unsafe for ramp-like IBR dynamics.

Result: RA-EKF, SOGI-FLL, Koopman, and TFT remain below 0.05 Hz across the three standard cases; EKF and IpDFT fail through intermittent ramp divergence.

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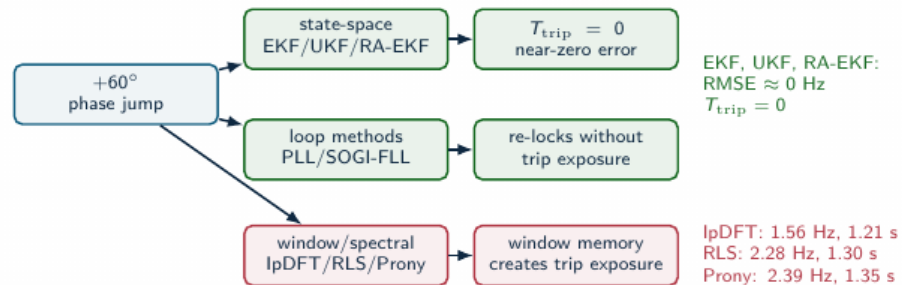
Standard scenario results (build-up)

Say:

- On the standard cases most methods are excellent, below fifty millihertz.
- CLICK: watch the ramp. The textbook EKF and IpDFT intermittently diverge, 37 and 14 hertz.

Key line: Robustness, not point accuracy, separates the families.

A clean phase jump is a family separator



The clean jump does not expose RA-EKF's main advantage. It shows which estimator families carry memory through discontinuities.

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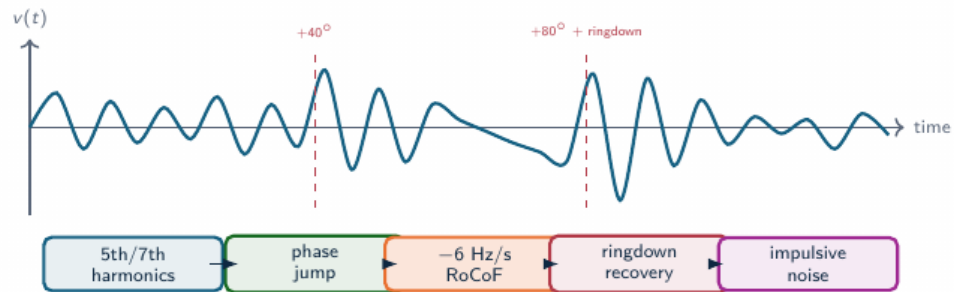
Scenario D: phase jump separates families

Say:

- A clean 60-degree phase jump separates the families.
- State-space filters and PLLs recover with zero trip exposure.
- Windowed and spectral methods carry over a second of exposure.

Key line: The phase jump does not break the EKF.

Composite IBR sequence: why it is hardest



Why standard tests miss it

The estimator sees spectral leakage, loop re-lock, state-model mismatch, transient recovery, and noise in one nonstationary waveform.

Why ATLAS decomposes it

The multi-event case proves the failure exists; severity sweeps identify which stressor actually breaks each estimator family.

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Composite IBR sequence: hardest case

Say:

- Scenario E stacks everything: harmonics, two phase jumps, a RoCoF segment, ringdown and impulsive noise.
- Spectral leakage, loop re-lock and model mismatch happen at once.

Key line: This is the hardest, most realistic case.

Scenario E: the leaderboard breaks

Method	RMSE [Hz]	T_{trip} [s]	CPU [μ s]
UKF	1.09	0.78	14
EKF	1.13	0.87	11
SOGI-PLL	1.14	0.78	11
SOGI-FLL	1.14	0.84	9
IpDFT	1.14	1.35	17
RA-EKF	1.16	0.92	15
Koopman	1.90	0.84	15 241
ZCD	426	1.55	7

No accurate winner

Under the stacked sequence every usable method collapses to ≈ 1.1 Hz RMSE; ZCD diverges catastrophically (426 Hz). High accuracy is simply not available here.

Source: full_mc_benchmark, $n = 5$: IBR_Multi_Event

Scalar ranking failure

UKF has the lowest RMSE and trip exposure, SOGI-FLL the lowest usable CPU, Koopman is 1000 $\times$ costlier; RA-EKF sits mid-pack but never diverges.

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Scenario E: the leaderboard breaks

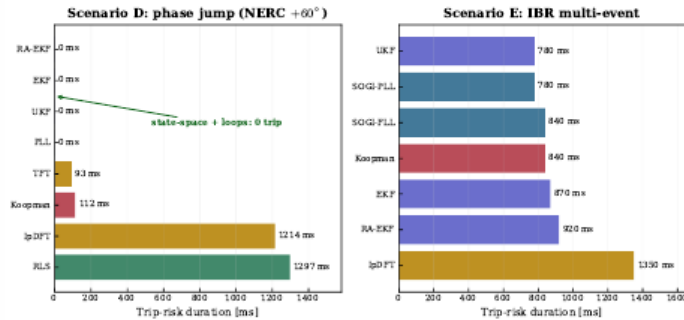
Say:

- Every usable method collapses to about 1.1 hertz RMSE.
- ZCD diverges to 426 hertz.
- RMSE, trip-risk and CPU rankings all disagree.

Key line: There is no accurate winner here.

Trip-risk evidence: core stress cases

Trip-risk is an event metric, not an average-error metric



Result reading

- Scenario D: state-space filters and PLL keep $T_{\text{trip}} = 0$; windowed/spectral methods (IpDFT, RLS) exceed 1.2 s.
- Scenario E: every method carries 0.8–1.4 s exposure — composite stress is unavoidable.
- The trip-exposure ranking is not the RMSE ranking.

Result: T_{trip} separates protection exposure from average error.

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Trip-risk evidence (figure)

Say:

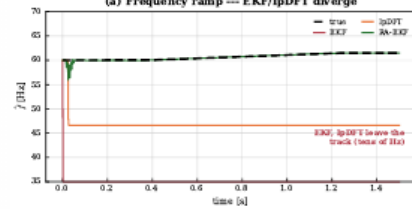
- In Scenario D the state-space filters keep zero exposure; windowed methods exceed 1.2 seconds.
- In Scenario E everyone carries exposure.

Key line: Trip-risk is not the RMSE ranking.

The failures are visible in the trace, not only in the table

RAMP DIVERGENCE

Event responses (full_mc_benchmark): intermit
(a) Frequency ramp --- EKF/IpDFT diverge

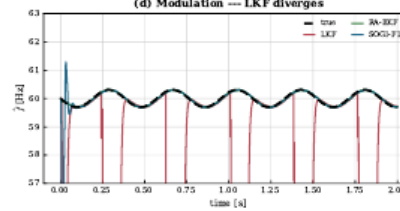


Failure signature

EKF and IpDFT leave the true ramp almost immediately; after the event, the estimate is no longer operationally usable.

MODULATION DIVERGENCE

(d) Modulation --- LKF diverges



Failure signature

LKF stays plausible between collapses, then repeatedly drops out of the usable frequency band.

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Event responses (figure)

Say:

- True frequency in black, each estimator tracking it.
- EKF and IpDFT visibly run away on the ramp; RA-EKF stays locked.

Key line: This is the intermittent divergence, seen directly.

Result comparison: winners depend on the question

Question	Best-supported answer	Evidence	Boundary
Clean step/modulation accuracy	State-space + SOGI-FLL near-zero; LKF2 best on mod	Step ≈ 0 ; mod 0.028–0.044 Hz	Clean cases do not predict ramp/event survival
Ramp and RoCoF tracking	RA-EKF is the most robust ramp tracker	0.006 Hz vs EKF 37 Hz, lpDFT 14 Hz (diverge)	Divergence is intermittent (seed-dependent)
Phase jump (Scenario D)	State-space filters recover with zero trip	EKF/UKF/RA-EKF $T_{\text{trip}} = 0$; lpDFT 1.21 s	Windowed/spectral carry the exposure
Multi-event RMSE	No accurate method — all ≈ 1.1 Hz	UKF 1.09 Hz best; ZCD 426 Hz diverges	RMSE hides trip-risk and 1000 $\times$ CPU spread
Embedded feasibility	PLL, SOGI-FLL, EKF, RA-EKF remain plausible	9–17 $\mu\text{s}/\text{sample}$; Koopman $\sim 15,000$	Python timing is a software-cost proxy

Selection changes with event family, objective metric, and compute envelope. A single leaderboard hides the deployment trade-off.

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Result comparison: winners by question

Say:

- Clean accuracy favors state-space and SOGI-FLL.
- Ramp robustness favors RA-EKF; multi-event has no winner; feasibility rules out the expensive methods.

Key line: Different question, different winner.

Computational feasibility

Method	Time/sample	Family	Target
ZCD	7 μ s	Loop	MCU
SOGI-FLL	9 μ s	Loop	MCU
PLL	9 μ s	Loop	MCU
EKF	11 μ s	Model	MCU/DSP
UKF	13 μ s	Model	DSP
RA-EKF	15 μ s	Model	DSP
IpDFT	17 μ s	Window	DSP/FPGA
ESPRIT	$\approx 1,400$ μ s	Window	CPU
Koopman	$\approx 15,000$ μ s	Data-driven	CPU/GPU

Source: full_mc_benchmark, $n = 5$: mean m13_cpu_time_us

Deployment interpretation

The Python baseline separates feasible estimator families from clearly prohibitive ones; hardware latency requires platform timing.

Key comparison

The state-space and loop families all run at ~ 10 – 17 μ s/sample (embedded-feasible); spectral (ESPRIT, MUSIC) and data-driven (Koopman) methods are 100 – $1000\times$ costlier and data-dependent.

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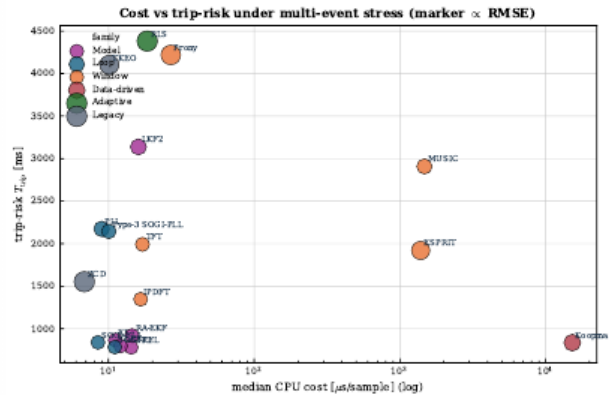
Computational feasibility

Say:

- Loop and state-space methods run in 9 to 17 microseconds per sample, embedded-feasible.
- Spectral and data-driven methods are 100 to 1000 times more expensive.

Key line: Cost alone removes several methods from a relay.

Deployment limits



Deployment interpretation

- Deployment combines RMSE, trip exposure, CPU, and divergence robustness.
- State-space + loop methods cluster at low cost ($\leq 17 \mu s$) and low trip.
- Spectral/data-driven methods trade lower error for 100–1000 $\times$ higher CPU.
- RA-EKF stays low-cost with no ramp divergence in this run.

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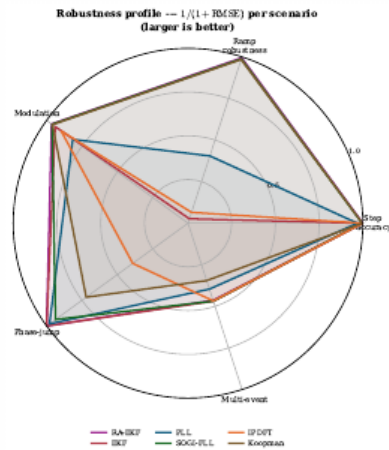
Deployment limits (figure)

Say:

- Deployment combines accuracy, trip exposure, cost and divergence-robustness.
- The low-cost cluster, state-space and loops, is also the low-trip cluster.

Key line: The practical winners cluster at low cost and low trip.

Balance profile



Radar profile

- Each axis is one scenario; larger means more robust ($1/(1 + RMSE)$).
- RA-EKF and SOGI-FLL stay robust across all five scenarios.
- EKF and IpDFT collapse on the ramp axis (intermittent divergence).
- No estimator is strong on every axis — robustness is the discriminator.

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Balance profile (radar)

Say:

- Each axis is one scenario; larger means more robust.
- RA-EKF and SOGI-FLL stay robust everywhere; EKF and IpDFT collapse on the ramp.

Key line: No method is strong on every axis.

Compliance heatmap: why standard tests are insufficient

Method	Step	Ramp	Mod	Jump	Multi
Koopman	P	P	P	F	F
TFT	P	P	P	F*	F
IpDFT	P	F	P	F	F
SOGI-FLL	P	P	P	P	F
PLL	P	F	F*	P	F
EKF	P	F	P	P	F
UKF	P	P	P	P	F
RA-EKF	P	P	P	P	F
LKF2	F*	P	P	P	F

Columns: **Jump** = phase jump (D), **Multi** = multi-event (E). **P** pass | **F** fails a declared RMSE/ T_{trip} /settling threshold | **F*** borderline.

Result: Step, ramp, and modulation compliance does not predict phase-jump or multi-event readiness.

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Compliance heatmap (build-up)

Say:

- Most methods pass step, ramp and modulation.
- CLICK: the same methods fail the phase-jump and multi-event columns.

Key line: Passing standard tests does not predict event readiness.

Empirical claims

Protection behavior

Divergence, not phase jumps, is the dominant failure mode.
On the ramp, EKF and IpDFT intermittently diverge (37 and 14 Hz mean); RA-EKF stays at 0.006 Hz. A clean phase jump is recovered by every state-space filter with $T_{\text{trip}} = 0$.

Standard-test limitation

Simple-case pass/fail leaves event readiness unresolved.
The heatmap shows methods that pass Step/Ramp/Mod but fail islanding or multi-event IBR stress.

Guardrail: report scenario, metric, and compute envelope before naming a preferred estimator.

Metric disagreement

RMSE, trip-risk, and CPU select different winners.
Under multi-event stress UKF has the lowest RMSE/trip, SOGI-FLL the lowest usable CPU, and Koopman is 1000× costlier.

Deployability

Feasibility changes the accuracy ranking.
RA-EKF runs at $\sim 15 \mu\text{s}/\text{sample}$; the spectral and data-driven methods are 100–1000× costlier and sit outside relay-class timing.

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Empirical claims

Say:

- Divergence, not phase jumps, is the dominant failure mode.
- Pass/fail leaves event readiness unresolved; the three metrics pick different winners; feasibility changes the ranking.

Key line: Four claims, one theme: report the conditions.

Expanded and ATLAS (section)

Expanded & ATLAS

Does the pattern scale across families and severity?

Key line: Does the pattern hold across families and severity? That is ATLAS.

Expanded stress analysis

FULL MONTE-CARLO BENCHMARK: 18 estimators, 33 scenarios, $n = 5$ per scenario

Monte-Carlo benchmark (full_mc)

- 18 estimators across 5 families.
- 33 scenarios (A–E plus severity variants).
- $n = 5$ Monte Carlo seeds per scenario.
- Standard, phase-jump, multi-event, and CPU metrics.

ATLAS severity sweeps

- Severity, sign, and policy curves per stress family.
- Fixed-policy, paper-grade $n = 30$ across all 9 required sweeps (both signs).
- Harmonic, RoCoF, phase-jump, and noise atlases.
- Tests sensitivity, scaling, and family stability.

Two evidence layers on the same truth: a replicated Monte-Carlo benchmark, plus ATLAS severity sweeps that trace where each estimator family degrades.

Expanded stress analysis

Say:

- We expand to a full Monte-Carlo benchmark: 18 estimators, 33 scenarios, five seeds.
- Plus the ATLAS severity sweeps.

Key line: We scale the test, not just repeat it.

Expanded benchmark: family winners

FULL MONTE-CARLO BENCHMARK: 18 estimators, 33 scenarios, $n = 5$ per scenario

Family	Count	RMSE leader	RMSE	Trip-risk leader
IEEE ramps	9	RA-EKF	11 mHz	0 s (state-space)
Magnitude steps	7	UKF	<1 mHz	0 s (state-space)
Modulation	3	LKF2	15 mHz	0 s (state-space)
Phase jumps	3	UKF	<1 mHz	0 s (state-space)
Frequency step	1	RA-EKF	52 mHz	RA-EKF, 3 ms
OOB interference	1	UKF	<1 mHz	0 s (state-space)
IBR harmonics	3	SOGI-FLL	61 mHz	SOGI-PLL, 10 ms
IBR ringdown	4	LKF2	202 mHz	ESPRIT, 87 ms
IBR multi-event	1	UKF	1.09 Hz	UKF, 780 ms

The full Monte-Carlo sweep confirms the central pattern: the RMSE leader rotates (RA-EKF, UKF, LKF2, SOGI-FLL) by disturbance family, and error grows from sub-mHz on clean cases to ~ 1 Hz under multi-event stress.

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Expanded benchmark: family winners

Say:

- The winner rotates: different methods lead the step, the ramp, harmonics and the multi-event.

Key line: The leaderboard is regime-dependent, not absolute.

ATLAS severity analysis

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)



A–E failure mode

When does error start to grow?

Are positive and negative events symmetric?

Does tuning policy change the rank?

Which results replicate?

ATLAS maps where estimator behavior changes with severity, sign, tuning policy, and replication depth.

Source: docs/ATLAS_PIPELINE.md; docs/ATLAS_METHOD_AUDIT.md

Say:

- ATLAS sweeps each stress by severity and by sign, fixed policy, n equals 30.
- This is now paper-grade across all nine required sweeps.

Key line: Paper-grade severity evidence, not a pilot.

ATLAS result: metric choice changes the answer

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)

Comparison	Different winner	Levels tested
RMSE vs trip-risk	83	113
RMSE vs RoCoF max error	55	113
RMSE vs RoCoF RMS error	50	113

Winner concentration

RMSE is led mostly by EKF and ESPRIT. Trip-risk is led mostly by Koopman, EKF, SOGI-FLL, ZCD, and ESPRIT. CPU is always led by ZCD.

Result: across 113 severity levels, the preferred estimator changes when the objective changes.

Source: atlas-papergrade-missing-v1/global_metrics_report.csv

Why this matters

For protection-adjacent use, a low average error is not enough. The estimator must avoid long deadband exposure, track RoCoF, and remain computable.

Operational implication

The benchmark output is not a leaderboard. It is a metric-conditioned selection map.

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ATLAS: metric choice changes the answer

Say:

- Across 113 severity levels, the preferred estimator changes when the objective changes.
- In 83 of them, RMSE and trip-risk disagree.

Key line: The output is a selection map, not a leaderboard.

ATLAS result: where estimators stop being valid

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)

Stress family	Maximum tested level	Methods below 0.05 Hz	Interpretation
Magnitude step	$\pm 90\%$ sag/swell	2	EKF and RA-EKF survive both signs.
AM modulation	10 Hz envelope	10	AM alone is not the hardest modulation case.
FM modulation	10 Hz modulation	0	FM is qualitatively harder than AM.
RoCoF ramp	± 50 Hz/s	0	Severe RoCoF requires a separate qualification class.
Phase jump	$\pm 60^\circ$	0	Single jump severity already breaks the 0.05 Hz guide.
Harmonics	30% THD	0	Low-THD winners do not survive extreme THD.
Interharmonics	20% at 75 Hz	0	Integer-harmonic tests are not sufficient.
Broadband noise	$\sigma = 0.1$ pu	0	Noise robustness is a first-class stress dimension.

ATLAS: where estimators stop being valid

Say:

- ATLAS tells us the severity at which each method crosses into failure.

Key line: That validity boundary is different for every family.

AM, FM, and interharmonics split the story

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)

Stress	Winner pattern	Threshold behavior
AM modulation	EKF wins slow, reference, fast, and severe regions.	Ten methods stay below 0.05 Hz through 10 Hz envelope modulation.
FM modulation	TFT leads slow; ESPRIT leads reference/fast/severe; SOGI-PLL reaches the best threshold.	No method stays below 0.05 Hz at 10 Hz FM modulation.
Interharmonics	EKF wins trace, reference, high, and severe levels.	EKF crosses the 0.05 Hz guide around 8%; most other families cross earlier.

Implication

AM, FM, integer harmonics, and interharmonics should not be collapsed into one “distortion” label. They select different estimator families.

Decomposition insight

A composite IBR trace can contain AM, FM, harmonics, interharmonics, noise, and phase steps. ATLAS isolates which stressor changes the ranking.

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AM, FM and interharmonics

Say:

- AM is easy and the model-based methods win.
- FM and interharmonics are harder, and the spectral methods take over.

Key line: Even within modulation, the winner switches.

When the textbook filter diverges

FULL MONTE-CARLO BENCHMARK: 18 estimators, 33 scenarios, $n = 5$ per scenario

Per-seed frequency RMSE [Hz] on the IEEE frequency-ramp scenario (5 Monte Carlo seeds):

Estimator	seed 1	seed 2	seed 3	seed 4	seed 5	trip-risk
EKF (textbook)	0.016	63.7	0.017	121.5	0.027	up to 1.35 s
RA-EKF	0.0055	0.0062	0.0083	0.0045	0.0058	0 s

What happens

The standard EKF tracks most seeds but **intermittently diverges** to tens of Hz. It is *not* a severity threshold: the worst seed (121 Hz) had a low ramp rate (1.7 Hz/s) while the steepest (5 Hz/s) tracked cleanly. The failure is triggered by onset/phase combinations — seed-dependent and silent.

Why RA-EKF matters here

Covariance scaling and event gating remove the divergence: stable on **every** seed, zero trip exposure. This — not a universal RMSE win — is its defensible value.

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When the textbook filter diverges

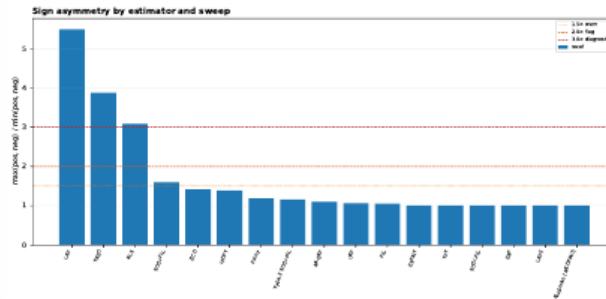
Say:

- The textbook EKF tracks most seeds but intermittently runs away to tens of hertz.
- It is not severity-driven; the worst seed had a low ramp rate.

Key line: The robustified RA-EKF removes it on every seed.

RoCoF has a sign: down-ramps are harder

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)



Source: atlas-papergrade-missing-v1/atlas_sign_asymmetry.csv ($n = 30$, both signs)

Directional bias (RMSE₋/RMSE₊ at ± 50 Hz/s)

- LKF 5.5 $\times$ worse on -RoCoF
- TKEO 3.9 $\times$, RLS 3.1 $\times$
- SOGI-FLL 1.6 $\times$
- EKF, UKF, RA-EKF ≈ 1.0 (symmetric)

Why it matters

A +RoCoF-only test can certify a method that fails on frequency *decline* — the protection-critical direction. Both signs must be swept.

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RoCoF has a sign

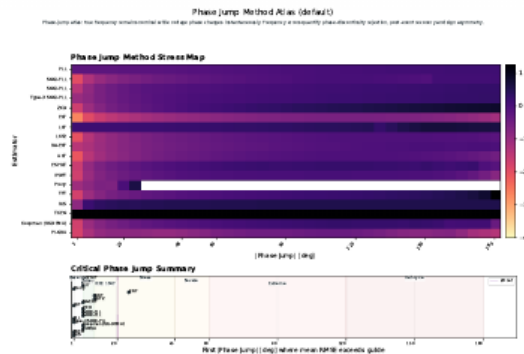
Say:

- Down-ramps are harder than up-ramps.
- LKF is 5.5 times worse on negative RoCoF, TKEO almost 4.

Key line: A positive-only test can pass a method that fails on frequency decline.

Phase-jump severity is non-monotone

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy



Result: Severity is non-monotone; sweep the jump angle rather than qualify one nominal point.

Worst case is not 180°

Most methods peak near 90–120° and recover toward 180° (a half-cycle jump partly self-cancels). EKF peaks at only 0.25 Hz @ 115°, back to 0.07 Hz @ 180°.

The TFT cliff

One exception: TFT rises gently, then jumps 2.5 → 21.9 Hz between 150° and 180° — a phase-reversal singularity.

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Phase-jump severity is non-monotone

Say:

- The worst case is not 180 degrees.
- Most methods peak around 90 to 120 degrees and then recover, because a half-cycle jump partly self-cancels.

Key line: Bigger is not always worse; sweep the severity.

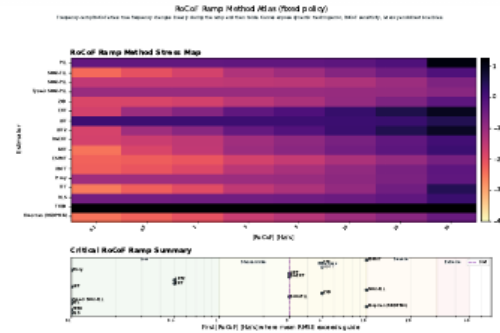
No estimator owns the whole stress space

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)

RoCoF regime	Best	med. RMSE
Low (0.1–1 Hz/s)	Koopman	4.1 mHz
Standard (1–3)	ESPRIT	7.6 mHz
IBR stress (3–10)	ESPRIT	22.3 mHz
Severe (10–30)	SOGI-PLL	70.4 mHz

Across stress types

EKF wins phase jumps, the magnitude step and interharmonics; ESPRIT/Koopman win the RoCoF and frequency-step regimes; ZCD wins low-THD harmonics; SOGI-PLL leads severe RoCoF. No method leads everywhere.



The deliverable is a selection map: event class $\times$ metric $\times$ compute budget.

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No universal winner (map)

Say:

- The champion rotates by RoCoF regime.
- Koopman at low rates, ESPRIT in the middle, SOGI-PLL when severe.

Key line: No universal winner; the deliverable is the map.

ZCD is a narrow instrument, not a general estimator

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)

Low-stress champion

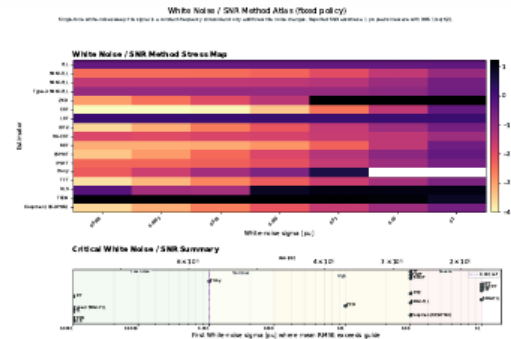
Up to ~15% THD, zero-crossing detection gives the *lowest* error of any method — **0.27 mHz** at 1% THD, 2.2 mHz at 15%; and just 1.1 mHz at low broadband noise.

Broadband-noise cliff

Same method: 34 mHz at $\sigma = 0.003$ pu $\rightarrow$ **3112 Hz** at $\sigma = 0.1$ pu. EKF stays ≤ 0.43 Hz across the same sweep.

Harmonic cliff

Same method again: 2.2 mHz at 15% THD $\rightarrow$ **39 Hz** at 20% $\rightarrow$ **86 Hz** at 30%.



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The ZCD paradox (build-up)

Say:

- Below 15 percent THD, ZCD is the most accurate method we have.
- CLICK: broadband noise pushes it to 3000 hertz.
- CLICK: above 20 percent THD it collapses to tens of hertz.

Key line: The same estimator is the best and the worst.

Voltage sags and swells are not symmetric

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)

Estimator at $\pm 90\%$	Sag to 0.1 pu	Swell to 1.9 pu
EKF RMSE	47 mHz	14 mHz
RA-EKF RMSE	49 mHz	30 mHz
ZCD RMSE	374 Hz	9.4 mHz
ZCD peak error	2856 Hz	32 mHz

Mechanism

Zero-cross timing error scales roughly as $\sigma_n/(A\omega)$. A deep sag lowers zero-cross slope and makes false crossings likely when absolute noise is fixed.

Source: atlas-papergrade-missing-v1/global_metrics_report.csv; src/estimators/zcd.py

What to claim

ZCD is amplitude-invariant only in the noise-free ideal case. Under fixed absolute noise, deep voltage sags create a low-SNR crossing problem.

Benchmark consequence

Sag and swell signs must be swept separately; a positive-only magnitude-step test can hide the worst case.

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Sags and swells are not symmetric

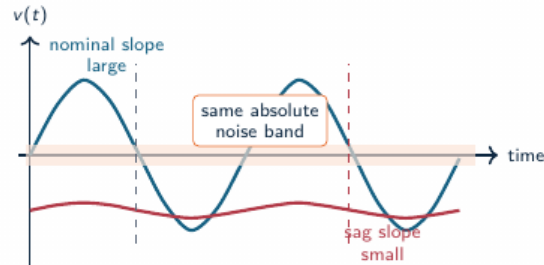
Say:

- A deep sag is far harder than a swell.
- ZCD goes from millihertz on a swell to hundreds of hertz on a sag.

Key line: Voltage direction matters; sweep both signs.

Why a voltage sag breaks zero-crossing detection

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)



Timing sensitivity

$$\Delta t \approx \frac{n(t_0)}{\frac{dv}{dt}|_{t_0}} \quad \frac{dv}{dt}|_0 \approx A\omega$$

When the voltage amplitude A collapses, the same noise produces much larger crossing-time error.

Estimator consequence

ZCD is elegant at clean high-SNR crossings, but deep sags convert amplitude loss into apparent frequency jitter and false crossings.

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Why a sag breaks zero-crossing

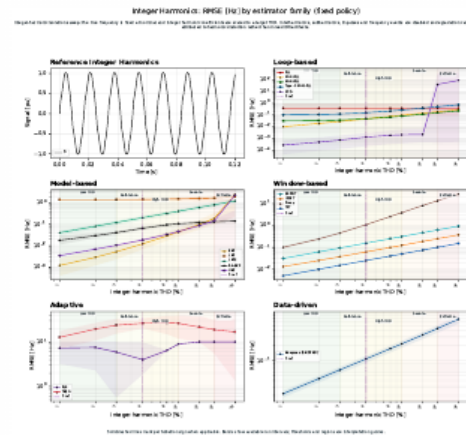
Say:

- Zero-cross timing error scales with noise over amplitude times frequency.
- A deep sag flattens the slope at the crossing, so noise creates false crossings.

Key line: The mechanism is physical, not a coding bug.

Harmonic degradation differs by estimator family

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)



Result

- Loop, window, model, and adaptive families degrade with THD at different rates.
- Best low-THD accuracy does not predict high-THD survival.
- Loop methods (ZCD) win at low THD but collapse above ~15%; window methods (TFT) degrade gently and lead at extreme THD.

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Harmonic degradation by family

Say:

- Loop methods like ZCD win at low THD but collapse above 15 percent.
- Window methods degrade gently and lead at extreme THD.

Key line: Low-THD accuracy does not predict high-THD survival.

CPU-accuracy Pareto

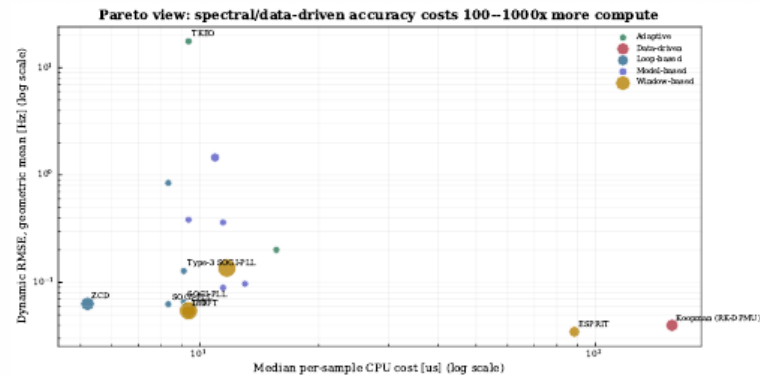
Say:

- Spectral and data-driven methods buy a little accuracy for 100 to 1000 times more compute.

Key line: For relay-class timing, that trade is usually not worth it.

Accuracy costs 100–1000× more compute

FULL MONTE-CARLO BENCHMARK: 18 estimators, 33 scenarios, $n = 5$ per scenario



Low-cost core
PLL, SOGI-FLL, EKF,
and RA-EKF sit in the
deployable compute
range.

Accuracy premium
Spectral and data-driven methods can move right by 100–1000× CPU cost.

Key line: So what do we actually deploy?

The slide features a dark blue background with a geometric pattern of overlapping triangles and lines. In the top right corner, there is a small blue logo for SGSMA 2026. The main title "Findings & deployment" is centered in a large, white, serif font. Below it, the subtitle "What to deploy — and what comes next" is centered in a smaller, white, sans-serif font. At the bottom left, the text "SGSMA 2026 · SANTIAGO DE CHILE" is displayed in white. Along the bottom edge, there are four logos: PES (Power & Energy Society), IAS (Instrumentation & Measurement Society), cigre (United Grids), and ics (International Council on Large Electric Systems).

Findings & deployment

What to deploy — and what comes next

SGSMA 2026 · SANTIAGO DE CHILE

PES
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SOCIETY

cigre
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ics

Three findings

1. Metric-conditioned

The RMSE leader often differs from the trip-risk and RoCoF leader across ATLAS severity levels.

2. Stress-bounded

AM remains survivable for many methods; severe FM, RoCoF, interharmonics, noise, and extreme THD break the 0.05 Hz guide.

3. Robustness over point accuracy

EKF, UKF, and IpDFT can diverge on ramps; RA-EKF keeps the ramp family stable across seeds.

Dynamic frequency estimators should be qualified by event family, sign, severity, recovery, trip exposure, and compute envelope.

Three findings

Say:

- One: a latency-versus-stress trade-off.
- Two: robustness beats point accuracy. RA-EKF's explicit RoCoF state stays stable where EKF and IpDFT diverge.
- Three: compliance tests are insufficient for IBR events.

Key line: Robustness, not accuracy, is the deployment discriminator.

Validity limits and next steps

Validation limits

- Limited to the reported scenario set.
- CPU is a software-level cost indicator; hardware latency requires platform timing.
- The study is simulation-only.
- Scenario-wise tuning can affect rank stability.

Validation program

- Hardware-in-the-loop validation.
- Wider converter-control interaction coverage.
- Sensitivity analysis of the tuning protocol.
- ATLAS severity and sign sweeps with validation gates.

These limits define the validation path: broader event coverage, hardware timing, fixed-policy replication, and field replay.

Validity limits and next steps

Say:

- Core means are n equals 5; ATLAS sweeps are paper-grade at n equals 30.
- Next is journal-grade at n equals 100, plus subspace and ML methods.

Key line: We are explicit about diagnostic versus paper-grade.

Recommendation map



Deployability depends on event class, trip-risk tolerance, and compute budget.

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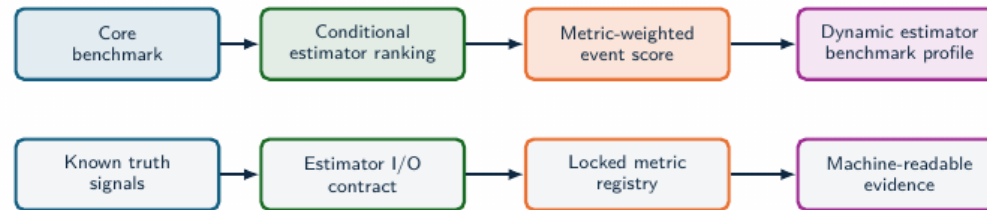
Recommendation map

Say:

- For fast frequency response and anti-islanding, RA-EKF is the strongest benchmark-backed choice.
- PLL and EKF are low-cost baselines; IpDFT and TFT where observation delay is acceptable.

Key line: A recommendation map, not a single ranking.

Qualification profile



Estimator selection moves toward **qualification** when each method is tested against a declared event profile, metric profile, and latency-cost envelope.

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Qualification profile

Say:

- A qualification profile tests transient recovery, trip exposure, compute feasibility and standard-case RMSE.

Key line: Qualify on all dimensions together, not one at a time.

Standardization profile

Current standardization layer

- PMU standards define phasor, frequency, and RoCoF measurement requirements.
- Compliance tests are necessary, with estimator-family ranking by IBR event survival left outside scope.
- Device reports often hide algorithmic failure modes behind aggregate accuracy.

Standardization boundary

A dynamic-estimator profile needs declared signals, estimator I/O, metrics, evidence artifacts, and latency-cost reporting.

Benchmark profile

- Canonical events: steps, ramps, phase jumps, ringdown, harmonics, interharmonics, noise, and composite IBR sequences.
- Locked metrics: RMSE, peak error, RoCoF error, trip-risk, settling, invalid samples, CPU, and latency.
- Task profiles: forensic monitoring, PMU augmentation, DFR support, and relay-adjacent screening.

Standardization value

A reference test profile needs the same signals, estimator I/O, metrics, and machine-readable evidence.

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Standardization profile

Say:

- We propose reporting these dimensions explicitly.
- So two estimators can be compared on equal terms.

Key line: Make the comparison reproducible and fair.

IBR Event Qualification Score

COMPOSITE EVENT SCORE: weighted event-class qualification metric

Score for estimator e

$$\text{IBR-ERS}(e) = 1 - \sum_{s \in \mathcal{S}} w_s \Phi_s(e),$$

$$\Phi_s(e) = \alpha \tilde{E}_s + \beta \tilde{P}_s + \gamma \tilde{T}_s + \delta \tilde{S}_s + \eta \tilde{C}_s + \lambda \tilde{L}_s.$$

Score composition

RMSE	peak	trip	settle	cost
$\tilde{E}_s, \tilde{P}_s$	dynamic and peak error			
$\tilde{T}_s, \tilde{S}_s$	trip exposure and recovery			
$\tilde{C}_s, \tilde{L}_s$	compute and latency penalties			

Why a score is needed

RMSE, trip-risk, transient recovery, and cost lead to different decisions. A composite score makes the deployment objective explicit.

Protection profile

Raise γ, δ, λ when trip exposure, recovery, and latency dominate the risk model.

Monitoring profile

Raise α, β, η when reconstruction accuracy and compute budget dominate.

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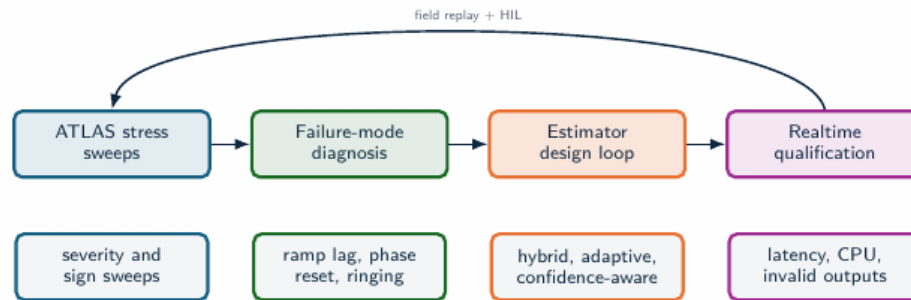
IBR Event Qualification Score

Say:

- A composite, weighted, event-class metric.

Key line: One number when you need one, without hiding the dimensions.

Toward realtime IBR estimators



- The stress analysis identifies which physical stressor breaks each estimator family.
- That diagnosis supports hybrid estimators: event classifiers, harmonic-aware preprocessing, adaptive bandwidth, and self-reported confidence.
- Validation should combine synthetic truth, EMT replay, hardware-in-the-loop, and field oscillography.

Say:

- Real-time estimators, Simulink microgrid fault records, modern and ML methods, C++ acceleration, ANDES co-simulation.

Key line: This is where the benchmark goes next.

Validation agenda and conclusion

Conclusion

- Field disturbances define the measurement problem; controlled synthetic truth makes estimator behavior explainable.
- The study compares estimators across objectives, regimes, and deployment constraints, including IBR-relevant stress modes.
- Empirical conclusions remain tied to evidence level, stress regime, and replication depth.

Validation agenda

- Full ATLAS sweeps: fixed policy, 100-run Monte Carlo, statistical rank-stability analysis.
- Composite qualification profiles for forensic, PMU-augmentation, and relay-adjacent tasks.
- Three-phase, unbalance, and converter-control interaction coverage.
- Real-event replay from DFR/PMU/oscillography with synchronized ground-truth proxies.
- Estimators that optimize the full objective vector: frequency error, RoCoF, trip exposure, settling, CPU, latency.

Result: estimator choice is tied to event class, metric target, and compute envelope; estimator qualification must declare the stress regime, latency limit, and evidence depth.

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Validation agenda and conclusion

Say:

- There is no universal frequency estimator.
- The defensible choice depends on the event, the metric and the compute budget.
- OpenFreqBench gives the community a reproducible way to make that choice.

Key line: Thank you. I am happy to take questions.